

WAB

Provadis School of International Management and Technology

Exposé

**Examining the Feasibility of Machine Learning-Based
LogP Prediction on an ESP32 Microcontroller**

A Proof-of-Concept Implementation and Performance Analysis

Rubin Chempananickal James

rubin.chempananickal-james@stud-provadis-hochschule.de

Department: Information Technology

Module: Embedded Systems and Software

Reviewer: Prof. Dr. Eric Hutter

February 16, 2026

Contents

Exposé	1
Problem Statement	1
Objectives	1
Methodology	1
Planned Structure	2

Exposé

Problem Statement

The accurate prediction of logP is crucial in various fields, including drug discovery and environmental chemistry. Traditional computational methods for logP prediction can be resource-intensive and may not be suitable for deployment on low-power embedded systems. This project aims to explore the feasibility of implementing machine learning-based logP prediction on an ESP32 microcontroller, which is widely used in IoT applications.

Objectives

The primary objectives of this project are:

- To develop a machine learning model for logP prediction using a suitable dataset.
- To optimize the model for deployment on an ESP32 microcontroller, considering constraints such as memory and computational power.
- To evaluate the performance of the implemented model in terms of accuracy and efficiency.

Methodology

The project will follow these steps:

1. Data Collection: Gather a dataset of chemical compounds with known logP values. The ZINC20 database will be used as a primary source for this data.
2. Model Development: Train a machine learning model (e.g., Random Forest, Support Vector Machine) using the collected dataset and the TFLite framework for microcontroller deployment.
3. Model Optimization: Use techniques such as quantization and pruning to reduce the model size for deployment on the ESP32.
4. Implementation: Deploy the optimized model on the ESP32 microcontroller and develop an interface for inputting chemical compound data.

5. Evaluation: Assess the model's performance using metrics such as mean absolute error (MAE) and inference time on the ESP32.

Planned Structure

The paper will likely be structured as follows:

- Introduction
- Literature Review
- Methodology
- Results and Discussion
- Conclusion and Future Work
- References
- AI Declaration
- Declaration of Authorship

Bibliography

- Baikété, Juda, Alhadji Malloum, and Jeanet Conradie (2026). „Comparative study of machine learning methods for accurate prediction of logP and pKb“. In: *Artificial Intelligence Chemistry* 4.1, p. 100101. ISSN: 2949-7477. DOI: <https://doi.org/10.1016/j.aichem.2025.100101>. URL: <https://www.sciencedirect.com/science/article/pii/S2949747725000181>.
- Irwin, John J. and Brian K. Shoichet (2005). „ZINC - A Free Database of Commercially Available Compounds for Virtual Screening“. In: *Journal of Chemical Information and Modeling* 45.1. PMID: 15667143, pp. 177–182. DOI: [10.1021/ci049714+](https://doi.org/10.1021/ci049714+). eprint: <https://doi.org/10.1021/ci049714+>. URL: <https://doi.org/10.1021/ci049714+>.